

# README.md

University Course Registration System  
Group 06 • Software Engineering

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## 1. Group Info

|              |                                       |
|--------------|---------------------------------------|
| Group Number | Group 06                              |
| Case Study   | University Course Registration System |
| Course       | Software Engineering                  |
| Instructor   | Samer Elkababji                       |

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|---------------------|------------|
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## 2. Overview

### System Purpose

This project is about building a web-based course registration system for a university. The idea is to give students a way to browse courses, sign up for them, drop or swap if needed, and check their grades and schedule — all in one place. The registrar handles things like opening and closing registration windows and managing how many students can join each course. Instructors can upload grades, see who's in their classes, and post announcements. The system also connects to external services for login, payment checks, and transcript generation.

### Tools Used

- Visual Studio Code — main development environment
- Draw.io — used for C4 diagrams, use case diagram, sequence diagrams, and activity diagram
- PlantUML — used for the class diagram and DFD
- GitHub — version control and team collaboration
- Pandoc — converting the report to PDF

### 3. Diagrams

| Diagram                         | Type          | Description                                                                  |
|---------------------------------|---------------|------------------------------------------------------------------------------|
| C4 Level 1 — Context Diagram    | C4 Model      | Shows who interacts with the system and what external systems it connects to |
| C4 Level 2 — Container Diagram  | C4 Model      | Breaks the system into its main parts: web app, server, and database         |
| Use Case Diagram                | UML           | Shows what each actor can do in the system                                   |
| Use Case Descriptions           | UML (Tabular) | A table going through each use case in detail                                |
| High-Level Sequence Diagram     | UML           | Shows how the main actors communicate with each other                        |
| Detailed Sequence Diagram       | UML           | Same as above but includes error cases and alternative flows                 |
| Class Diagram                   | UML           | Shows the structure of the system — classes, attributes, and relationships   |
| Activity Diagram with Swimlanes | UML           | Shows step-by-step workflows across different actors                         |
| DFD — View Academic Records     | DFD           | Shows how data flows when a student requests their academic records          |

### 4. Repository Structure

| Folder / File    | Description                                 |
|------------------|---------------------------------------------|
| /docs/report.md  | Report written in Markdown                  |
| /docs/report.pdf | Final PDF version of the report             |
| /uml/*.puml      | PlantUML source files                       |
| /uml/*.drawio    | Draw.io source files                        |
| README.md        | This file — group info and project overview |

### 5. Contributions

| Member       | Role & Contributions                                                                          | Commits |
|--------------|-----------------------------------------------------------------------------------------------|---------|
| Lily Alalami | Report writing, PPT, sequence diagram explanations, high-level and detailed sequence diagrams | 32      |

| Member              | Role & Contributions                                                                                                 | Commits |
|---------------------|----------------------------------------------------------------------------------------------------------------------|---------|
| Nour Shalab         | C4 Level 1 & 2 diagrams, DFD diagram and description, C4 and DFD explanations                                        | 31      |
| Abdullah Mashharawi | Class diagram, activity diagram with swimlanes, activity and class diagram explanations                              | 12      |
| Rawan Abdo          | Use case diagram, use case description tables, generalization and aggregation on class diagram, related explanations | 28      |

GitHub Repository: <https://github.com/abdullah-mashharawi/Group-6-Software-Engineering>

# System Description

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## 1.1 Introduction

The University Course Registration System is a web-based platform built to handle the course enrollment process for students, instructors, and the registrar's office. Instead of dealing with manual paperwork or disconnected systems, everything is managed in one place — from browsing courses to generating transcripts. The goal is to make the process faster, more consistent, and easier for everyone involved.

## 1.2 What the System Does

At its core, the system lets students manage their academic schedule. They can look through available courses, register for ones they want, drop courses they no longer need, or swap one course for another. The system checks a few things before letting a student register — do they meet the prerequisites, is there still space in the course, is the registration window open, and do they have any financial holds. All of these checks happen automatically before anything gets confirmed.

On the admin side, the registrar controls when registration opens and closes, and sets the maximum capacity for each course. Instructors can log in to see who's in their classes, upload grades, and post announcements for their students. When a student needs an official transcript, the system sends a request to an external transcript service that generates and delivers it.

## 1.3 Actors

There are four main actors in this system:

- Student — the main user. Can browse courses, register, drop, swap, view their schedule, and check their grades.
- Instructor — can manage their course roster, submit grades, and post announcements.
- Registrar — handles the administrative side. Sets registration timelines, manages course capacities, and oversees enrollment policies.
- External Systems — three outside services the system talks to: the Authentication System (handles login), the Finance System (checks if a student has any payment holds), and the Transcript System (generates official transcripts).

## 1.4 Core Features

- Login and session management — all users log in through the institutional authentication service. Sessions are checked before any protected action is allowed.
- Course browsing — students can view the full list of available courses, including details like section info, timings, and seat availability.
- Registration — students register for courses after the system checks prerequisites, capacity, financial status, and whether registration is open.

- Drop and swap — students can remove a course or replace it with another, as long as it's within the deadline. Swapping combines both a drop and a new registration.
- Schedule and grade viewing — students can check their current schedule and any grades that have been uploaded.
- Grade submission — instructors submit grades, which get validated and stored.
- Roster management — instructors can view and manage their enrolled student list.
- Announcements — instructors can post updates that all students in that course can see.
- Transcript generation — students can request an official transcript, which is generated by the external transcript service.

## 1.5 System Architecture

The system follows a standard three-tier web architecture. The front end is a web application that all users interact with through their browser. Behind that is an application server that handles all the logic — checking prerequisites, enforcing deadlines, validating payments, and so on. Then there's a relational database that stores everything: student records, courses, enrollments, grades, and more.

The application server is also the only part that talks to the external systems. So when a student logs in, the server contacts the authentication service. When they register, it checks with the finance system. When a transcript is needed, it sends the request to the transcript service. Keeping all of that in one place makes the system easier to maintain and update.

## 1.6 Data

The database holds all the core data: users (students, instructors, registrars), courses, sections, enrollments, prerequisites, grades, schedules, announcements, and transcripts. The application server enforces all rules before writing anything to the database, so data stays consistent.

## 1.7 External Integrations

The three external systems — Authentication, Finance, and Transcript — are connected through the application server using defined interfaces. This means if the university decides to change one of those services, only the server-side integration needs to be updated, not the whole system. It keeps things flexible without breaking what already works.

# Part I: Context

## C4 Level 1 — Context Diagram

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The context diagram gives the highest-level view of the system — basically, who's involved and what outside services it talks to. There's no internal detail here, just the big picture of where the system sits within the university environment.

### Actors

- Student — uses the system to browse courses, register, drop or swap, and view their schedule and grades.
- Instructor — logs in to manage their roster, submit grades, and post announcements.
- Registrar's Office — the admin side. Sets enrollment rules, manages timelines, and monitors registrations.

### External Systems

- Authentication System — verifies who's logging in before granting access.
- Finance System — checks whether a student has any unpaid holds before allowing enrollment.
- Transcript System — receives academic data and produces official transcripts when requested.

The diagram makes it clear that the system sits in the middle — every interaction between a human actor and an external service goes through it. Nothing talks directly to the external systems except the core platform.

## C4 Level 2 — Container Diagram

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The container diagram goes one level deeper and breaks the system into its actual deployable components. This is where you can see how the different parts of the system are structured and how they communicate.

### Containers

- Web Application — the front-end interface built with HTML, CSS, and JavaScript. This is what students, instructors, and the registrar see and interact with. It sends requests to the application server and displays the responses.

- Application Server — the core of the system. All business logic lives here — enrollment rules, prerequisite checks, scheduling, grading policies. It also handles all communication with external systems.
- Relational Database — stores everything: students, courses, sections, enrollments, prerequisites, grades, announcements, schedules.

## External System Communication

- Authentication System — the server sends credentials here during login and session checks.
- Finance System — called before enrollment to confirm no financial holds exist.
- Transcript System — receives export requests and generates the official academic transcript.

The structure is a clean three-tier setup. The web app doesn't talk to the database directly, and it doesn't touch external systems. Everything goes through the application server, which acts as the single point of control.

## Part II: Interactions

### Use Case Diagram

The use case diagram gives an overview of the system's functionality from the user's perspective. It maps out which actors are involved in which operations, and shows some of the relationships between use cases.

#### Actors

- Student — initiates all enrollment-related actions and academic queries.
- Instructor — manages course content, grades, and announcements.
- Registrar — handles administrative controls like timelines and capacities.
- External Systems — participates in login and can query grades.

#### Relationships Between Use Cases

- Register for Course includes Verify Prerequisites and Check Capacity — both are mandatory sub-steps that run automatically every time a student tries to register.
- Swap Course includes both Drop Course and Register for Course — it's essentially both operations combined into one.
- View Schedule includes Browse Courses — you can't build a schedule view without pulling course data first.
- Post Announcements extends Upload Grades — announcements can be triggered as a follow-up to a grade upload.

The diagram groups use cases by actor responsibility which makes it easy to see at a glance what each person in the system can actually do.

### Use Case Descriptions

The table below goes through each use case in more detail — who's involved, what data is needed, what triggers it, and what the system does in response.

| Use Case | Actors                         | Description               | Data               | Stimulus                | Response             | Comments                       |
|----------|--------------------------------|---------------------------|--------------------|-------------------------|----------------------|--------------------------------|
| Login    | Student, Instructor, Registrar | User logs into the system | Username, Password | User enters credentials | System validates and | Common authentication use case |



| Use Case             | Actors                    | Description                                 | Data                                           | Stimulus                      | Response                                                              | Comments                                       |
|----------------------|---------------------------|---------------------------------------------|------------------------------------------------|-------------------------------|-----------------------------------------------------------------------|------------------------------------------------|
|                      | r,<br>External<br>Systems |                                             |                                                |                               | grants/denies access                                                  |                                                |
| Browse Courses       | Student                   | View available courses                      | Course list                                    | Student opens course catalog  | System displays all available courses                                 |                                                |
| Register for Course  | Student                   | Student enrolls in a course                 | Student ID, Course ID, Prerequisites, Capacity | Student selects a course      | System runs prerequisite and capacity checks then enrolls the student | Includes: Verify Prerequisites, Check Capacity |
| Verify Prerequisites | System                    | Checks if student meets course requirements | Student record, course requirements            | Triggered during registration | System validates eligibility                                          | Included use case for registration             |
| Check Capacity       | System                    | Checks if course has available seats        | Course ID, enrollment count, max capacity      | Triggered during registration | System allows or rejects registration                                 | Included use case for registration             |
| Drop Course          | Student                   | Student removes a course                    | Student ID, Course ID                          | Student selects drop option   | System removes student from course                                    |                                                |
| Swap Course          | Student                   | Replace one course with another             | Student ID, Old Course ID, New Course ID       | Student selects swap option   | System drops old course and registers new one                         | Includes Drop Course + Register for Course     |
| View Schedule        | Student                   | Display student timetable                   | Enrollment data                                | Student requests schedule     | System shows weekly schedule                                          |                                                |
| View Grades          | Student, External Systems | View academic grades                        | Student ID, Grades                             | User requests grades          | System displays grades                                                | External systems may query grades              |
| Manage Course Roster | Instructor                | Manage enrolled students                    | Course ID, Student list                        | Instructor updates roster     | System updates course enrollment list                                 |                                                |
| Upload Grades        | Instructor                | Enter student grades                        | Course ID, Grades                              | Instructor submits grades     | System stores grades                                                  | Must validate data format                      |

| Use Case                     | Actors     | Description                  | Data                         | Stimulus                   | Response                         | Comments |
|------------------------------|------------|------------------------------|------------------------------|----------------------------|----------------------------------|----------|
| Post Announcements           | Instructor | Publish course announcements | Announcement text, Course ID | Instructor posts update    | System publishes announcement    |          |
| Manage Course Capacity       | Registrar  | Set maximum seats per course | Course ID, Capacity limit    | Registrar updates capacity | System updates course limits     |          |
| Manage Registration Timeline | Registrar  | Control registration period  | Start date, End date         | Registrar sets timeline    | System opens/closes registration |          |

## High-Level Sequence Diagram

This diagram shows how the main actors communicate with each other during the system's key operations. It's meant to give a stakeholder-level view — the focus is on the overall flow rather than technical details or error handling.

### Authentication

The student sends a login request to the system. The system forwards this to the Institutional System for credential verification. Once cleared, the student gets a login success message.

### Student Enrollment

After logging in, the student initiates an enrollment request. The system checks with the Finance System to confirm there are no holds. If everything is clear, enrollment is confirmed.

### Course Browsing

The student requests the course list. The system returns all available courses.

### Registration

The student sends a registration request to the Registrar's Office. The registrar verifies the student's eligibility and tells the system to save the enrollment. Once that's done, the student gets a registration confirmed message.

### Drop or Swap

The student sends a drop or swap request. The registrar instructs the system to update the record. After the update, the student gets a confirmation.

## Academic Schedule and Grades

The student can request their schedule or grades at any time. The system retrieves and returns the relevant data.

## Instructor Operations

The instructor uploads grades, which the system saves. They can also request the course roster or post an announcement — both go through the system and get stored in the database.

## Transcript

The student requests a transcript. The system tells the Institutional System to generate it. Once it's ready, the transcript gets delivered back to the student.

## Detailed Sequence Diagram

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This version of the sequence diagram is aimed at developers. It covers the same flows as the high-level diagram but adds alternative paths, edge cases, and what happens when something goes wrong.

### Login — What if credentials are wrong?

When the student logs in, the system checks credentials with the Institutional System. If they're valid, an authentication token is issued. If not, the system returns an invalid credentials error and access is denied.

### Enrollment — What if there's a financial hold?

After login, the student tries to enroll. The system checks with the Finance System. If no hold exists, enrollment goes through. If a hold is detected, the system returns a hold error and the enrollment is blocked.

### Registration — Validation Chain

When the registration window is open, the system runs checks in sequence. First, prerequisites — if not met, the registration fails right there. Then capacity — if the course is full, it's rejected. Only if both pass does the enrollment get confirmed.

## **Drop and Swap — Deadline and Conditions**

For a swap within the deadline: the system checks prerequisites and capacity for the new course. If both are fine, the old course is dropped and the new one is added. If not, the swap is rejected. For a regular drop: the enrollment record is removed and the seat count goes back up. If the deadline has already passed, the system returns a period closed message and nothing changes.

## **Grades — What if they haven't been uploaded yet?**

When a student requests their grades, the system checks if the instructor has submitted them. If yes, they're returned. If not, the system returns a grades pending response.

## Part III: Structure

### Class Diagram

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The class diagram shows the full structure of the system — all the classes, what they contain, and how they relate to each other. It uses five types of relationships: generalization, aggregation, composition, association, and dependency.

#### Abstract Classes and Inheritance

The system uses abstract classes to avoid repeating the same code across similar entities. The `User` class is the parent of `Student`, `Instructor`, and `Registrar` — they all share things like a user ID, name, email, and the ability to log in and validate a session. The three external service classes (`AuthenticationSystem`, `FinanceSystem`, `TranscriptSystem`) all inherit from `SystemComponent`. Academic record types like `Transcript`, `Grade`, and `Enrollment` come from `AcademicRecord`. `Course` and `Section` both extend `CourseOffering`.

#### Aggregation and Composition

A `Course` aggregates multiple `Sections`, meaning a course can have several scheduled offerings and they exist independently of each other. A `Transcript` aggregates multiple `Grades` — it's made up of individual grade entries. The only composition in the diagram is between `Section` and `Announcement`. An announcement can't exist on its own; it belongs to a specific section. If the section goes away, so does the announcement.

#### Associations and Dependencies

The `Enrollment` class sits between `Student` and `Section` — it's the record that says a student is in a particular course offering. A `Student` is also linked to a `Schedule` that tracks their current timetable. The `Student` class depends on both `AuthenticationSystem` and `FinanceSystem`, which reflects the fact that login and financial checks have to happen before enrollment. `Course` has a self-association to handle prerequisites — a course can require another course.

## Part IV: Behavior

### Activity Diagram with Swimlanes

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The activity diagram shows how the system actually behaves when users interact with it. It uses swimlanes to separate what each actor or system is responsible for, and decision nodes to show where the flow can branch.

#### Login and Entry

Everything starts with the student logging in through the Authentication System. Once authenticated, a decision node lets them choose what to do next — register, drop, swap, or view academic info.

#### Registration

For registration, the system first checks the session is still valid. If not, the student has to log in again. Then it checks with the Registrar whether the registration window is open. If closed, the process stops. If open, the system runs three checks in order: prerequisites (Registrar), seat capacity (database), and financial hold (Finance System). Fail any one and registration ends. Pass all three and the system uses a fork node to do three things at once — update the academic record, update the course roster, and adjust seat count. Once those all finish, the student gets a confirmation and their updated schedule.

#### Drop and Swap

Dropping a course removes the student's enrollment record and frees up a seat. These two things happen in parallel. The student then sees their updated schedule. Swapping combines a drop and a registration — the old course is removed first, then the system tries to register the new one with all the same checks. If it works, everything is updated. If not, the student gets an error.

#### Viewing Academic Info

If the student just wants to see their schedule or grades, the system retrieves the data from the database — no prerequisite or financial checks needed. The schedule and grade retrieval can happen at the same time, and the result is displayed to the student.

#### Instructor Side

Instructors log in and can do three things: manage the roster, upload grades, or post an announcement. Uploading grades writes directly to the database. Posting an announcement stores it so enrolled students can see it.

## Data Flow Diagram — View Academic Records

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This DFD focuses specifically on what happens behind the scenes when a student requests their academic records. It shows how data moves through the system across five processes.

- Process 5.1 — Retrieve Student Profile: When the student sends a request, this process pulls their profile from the User Database using their student ID.
- Process 5.2 — Retrieve Enrollment History: The student ID is passed along to this process, which queries the Enrollment Database and returns all enrollment records for that student.
- Process 5.3 — Gather Grade Records: Using the enrollment data from the previous step, this process fetches the matching grade records from the Grades Database.
- Process 5.4 — Compile Academic History: The enrollment and grade data get combined here into a single complete academic history that gets passed forward.
- Process 5.5 — Generate and Export Transcript: The compiled history is formatted into an official transcript and sent to the external Transcript System for delivery.

The flow is straightforward — each process takes an input, does something with it, and passes the result to the next step. There are no loops or circular dependencies, which keeps the data flow clean and easy to follow.

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### GitHub Repository

<https://github.com/abdullah-mashharawi/Group-6-Software-Engineering>